Eur. J. Biochem. *146*, 475 – 480 (1985)

Perinatal development of the liver in rat and spiny mouse

Its relation to altricial and precocial timing of birth

Wouter H. LAMERS, Piet G. MOOREN, André DE GRAAF, and Robert CHARLES

Department of Anatomy and Embryology, Academic Medical Centre, University of Amsterdam

(Received August 3, 1984) – EJB 84 0854

Rat (*Rattus norvegicus*) and spiny mouse (*Acomys cahirinus*) are closely related murinoid species that mainly differ in the developmental timing of birth. A comparison between the developmental profiles of some characteristic enzymes of the liver of both species was carried out to elucidate the question to what extent are these enzymic profiles and hence the maturation of the liver related to the timing of birth? It was found that these organotypic enzymes first become detectable at the same developmental stage in both species. Likewise, the weaning phase of the enzymic profiles occurs at the same developmental time point in both species. It is argued that both the first appearance and the weaning increase in enzyme activity levels occur at endogenously programmed timepoints with only superimposed effects of hormones. In contrast, the perinatal phase of the enzymic profile is completely dependent on the developmental timing of birth and therefore appears not to be anchored to a particular developmental timepoint, but rather to be dependent on birth-associated (hormonal) adaptation. In accordance with this hypothesis it was found that the morphological development of the liver proceeded independent of the timing of birth. Furthermore, the hormonal regulation of the investigated enzymes was found to be the same in both species. Despite the more advanced state of morphological development of the liver in the spiny mouse at birth, it was found that the inducibility of organotypic gene expression by hormones in spiny mouse fetuses was as limited as in rat fetuses. This observation therefore suggests that the intra-uterine environment is responsible for the limited inducibility of enzymes before birth.

The enzymic development of the parenchymal cells of the rat liver is characterized by a cluster of enzymes that show their maximal activity in the perinatal period and by a cluster of enzymes that show steeply rising activities in the 3rd postnatal week [1 – 3]. These enzyme clusters are coordinated with similar clusters in other regions of the alimentary tract and in the respiratory tract [4, 5] and were designated as the perinatal and preweaning clusters respectively [1]. The common developmental profiles, that are represented by these clusters, point to a common control mechanism.

Experiments that were aimed at the elucidation of these control mechanisms have generally involved the manipulation of the hormonal status of the animals and have identified glucocorticosteroid hormone, cyclic AMP and, to a lesser extent, thyroid hormone as the main modulating factors of hepatocyte-specific gene expression [2, 3, 6 – 9]. However, these modulating effects of hormones are probably superimposed upon an inherent developmental program as ablation of, for example the steroid-hormone-producing glands attenuates, but does not abolish the perinatal and preweaning increases in enzyme levels [3, 10 – 13].

When addressing the pressing question: to what extent is the 'rat model' applicable to man? Greengard found that the sequence in which different enzymes approach their adult concentration in human liver resembles that in rat liver and suggested that the regulatory mechanisms responsible for gene expression are also analogous [14]. However, quantitatively the enzymic composition of human fetal liver at midgestation is very similar to that of rat liver just prior to birth and very little additional maturation during the remaining half of pregnancy is noted [14].

These observations point to the fact that rat fetuses start to prepare for birth biochemically and physiologically as soon as organogenesis is completed [15], thus providing a clear example of an altricial mode of development in mammals. In contrast, a precocial mode of development is characterized by a relatively long interval between the completion of organogenesis and the moment of birth. To further elucidate and analyse the factors that control the level and extent of gene expression of liver proteins in the fetal and neonatal period it could be very advantageous to complement the altricial 'rat model' with a model exhibiting the characteristics of a precocial mode of development. Ideally, this 'model' species should be closely related to the rat and differ in the developmental timing of birth only.

The spiny mouse, *Acomys cahirinus*, seems to meet this requirement, having a long gestation period (39 days), associated with small (2 – 4) litter size. Organogenesis proceeds somewhat slower in this species than in the rat, requiring one additional week to be completed [5]. The neonates show an advanced stage of development at birth, having a fur coat, open eyes and ears, and are capable of locomotion and thermoregulation. Self-feeding begins a few days after birth. The rat has developed to a comparable extent in the middle of the 2nd and 3rd postnatal week respectively. The late intra-

Enzymes. Glutamate dehydrogenase (EC 1.4.1.2); ornithine transcarbamoylase (EC 2.1.3.3); tyrosine aminotransferase (EC 2.6.1.5); glucose-6-phosphatase (EC 3.1.3.9); arginase (EC 3.5.3.1); carbamoyl-phosphate synthetase (ammonia) (EC 6.3.4.16).

uterine developmental stages of the spiny mouse, therefore, seem comparable to the early extra-uterine developmental stages of the rat [5, 16, 17]. Comparison of the developmental profiles of the rat and spiny mouse liver-specific enzymes may then answer the question of to what extent the developmental profiles are related to the developmental timing of birth. Furthermore the assessment of the effects of hormonal treatment on enzyme levels in the late fetal period may shown whether the incomplete induction of organotypic gene expression by hormones, that is seen in fetal rats [2, 6, 7], has a developmental or an environmental basis, i.e. whether it is due to an as yet uncompleted maturation of the parenchymal cells or to the intra-uterine stay. To that end we have determined the levels of six enzymes as well as protein and DNA content in the liver of fetal and neonatal spiny mouse as well as the effects of hormone treatment upon these parameters. These parameters were not chosen on the basis of specific functional importance, but in the hope of encompassing a spectrum of 'developmental behaviours' [14], including enzymes with perinatal and weaning developmental maxima in the rat [2, 8].

Part of this work was presented in preliminary form at a conference on Urea Cycle Diseases [5].

MATERIALS AND METHODS

Animals

Specimens of the spiny mouse (*Acomys cahirinus*) were obtained from our laboratory colony. The animals were fed pelleted rat chow (Hope Farms diet R.M.H.-B.). Fetal age was read from a chart relating body weight to gestational age [16].

Hormones were administered daily (prednisolone, 50 µg × g body weight^{-1}; triiodothyronine, 0.8 µg × g body weight^{-1}) or twice daily (glucagon, 25 µg × g body weight^{-1}) by intraperitoneal injection to newborn animals (Table 1). These doses were found previously to result in maximal stimulation of liver enzymes in rats [2, 6, 8]. Unfortunately, it was not possible repeatedly to administer hormones directly to the spiny mouse fetuses (cf. 2, 6–8) as the dams did not survive the necessary ether anaesthesia. Instead, dams were injected subcutaneously with 30 µg dexamethasone acetate × g body weight^{-1} for four consecutive days. On the last day the fetuses were exposed by caesarian section and injected intraperitoneally with 10 µg dexamethasone phosphate and 50 µg glucagon-Zn-protamine complex × g body weight^{-1} (Table 2). Glucagon-Zn-protamine complex was a gift from NOVO (Copenhagen, Denmark).

PREPARATION OF TISSUES AND ASSAY PROCEDURES

Tissue preparation

Animals were killed by decapitation and the liver was removed, weighed, frozen in liquid N_2 and stored at −80°C until assay. For preparation of enzyme extracts, the liver was homogenized in 9 volumes of 0.25 M sucrose. For the assay of carbomyl-phosphate synthetase I, ornithine transcarbamoylase and arginase, a 0.1% cetyltrimethyl-ammonium bromide, 1 mM dithiothreitol extract was prepared [2, 3]; for the assay of glucose-6-phosphatase and glutamate dehydrogenase, a 0.1% Triton X-100, 1 mM dithiothreitol extract was prepared [3]; and for the assay of tyrosine aminotransferase an enzyme extract was made by centrifuging an aliquot of the sucrose homogenate for 15 min at 30000 × g_{av}. Protein and DNA were estimated in the sucrose homogenate.

Assays

Carbamoyl-phosphate synthetase (ammonia) I, ornithine transcarbamoylase and arginase were estimated as described previously [2, 3]. Tyrosine aminotransferase activity was determined as described by Diamondstone [18]; glutamate dehydrogenase activity as described by Schmidt [19]; glucose-6-phosphatase activity as described by Baginsky et al. [20]; protein as described by Cleland and Slater [21] and DNA as described by Burton [22]. All enzyme activities were measured at 25°C except glutamate dehydrogenase, which was measured at 37°C. 1 unit enzyme activity is defined as the amount of enzyme that produces 1 µmol product × min^{-1} under the conditions of assay. Under these conditions all enzyme activities are linearly proportional to incubation time and protein concentration in the range used in this study.

In the charts, means and standard errors are depicted, representing 3–6 individual estimates. In Table 1 the minimum difference in average enzyme activity between two treatments, that is significant at the 5% level, was calculated according to Dunnett [23].

Histological procedures

For microscopical examination of the livers, dams were killed on the day of expected delivery. The fetuses were quickly removed, fixed in Bouin's fixative and embedded in paraplast. Sections (7 µm) were stained with haematoxylin and azophloxin.

RESULTS

Fig. 1 shows the developmental changes in enzyme activity of several liver enzymes in the spiny mouse that belong to the perinatal and weaning cluster in the rat. In all cases enzyme activity can be detected 12 days before birth, when the animals are developmentally comparable to rat fetuses 4 days before birth [5]. Subsequently two types of developmental profiles are seen: those of enzymes that attain their developmental maximum shortly after birth (tyrosine aminotransferase and glucose-6-phosphatase) and those of enzymes that attain their developmental maximum during the 2nd postnatal week (the urea cycle enzymes: carbamoyl-phosphate synthetase, ornithine transcarbamoylase and arginase).

On closer inspection it can be seen that the urea cycle enzymes, carbamoyl-phosphate synthetase, ornithine transcarbamoylase and arginase, show a rise in activity during the last 7–8 prenatal days up to approximately 10–50% of the levels attained at weaning. During the first 4 postnatal days enzyme activity does not increase. This plateau is followed by an increase to maximal levels during the next 10 days. These levels do not differ substantially from the adult values.

Tyrosine aminotransferase and glucose-6-phosphatase represent the group of enzymes that attain a developmental maximum shortly after birth. Tyrosine aminotransferase levels before birth are very low (less than 10% of those immediately after birth) and may increase slightly during the last 2 prenatal days. Glucose-6-phosphatase activity increases, like the urea cycle enzymes, during the last prenatal week up to approximately 25% of postnatal levels. In the 2nd postnatal week activities of tyrosine aminotransferase and glucose-6-

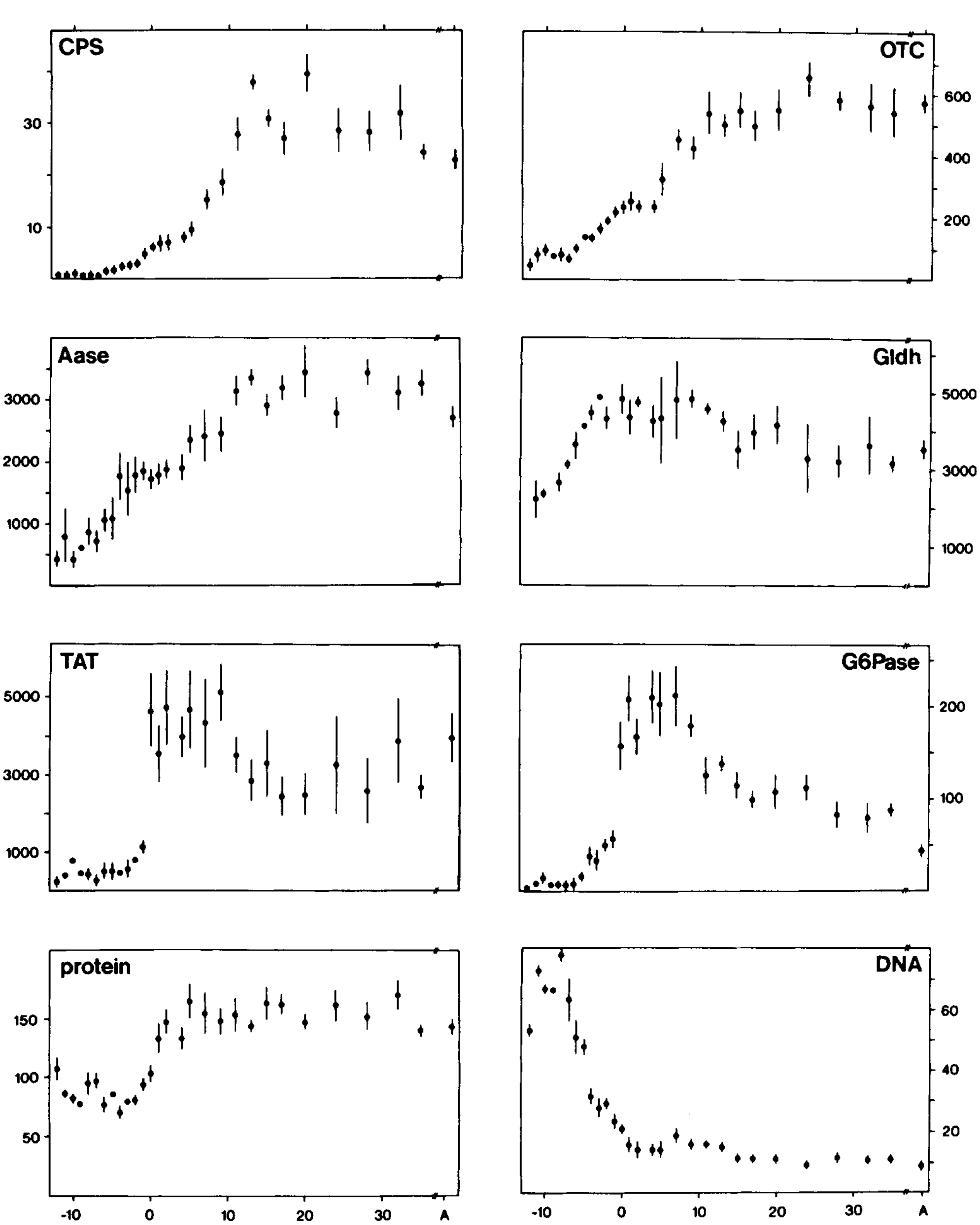

Fig. 1. *The developmental changes in the activity of carbamoyl-phosphate synthetase (CPS), ornithine transcarbamoylase (OTC), arginase (Aase), glutamate dehydrogenase (Gldh), tyrosine aminotransferase (TAT), glucose-6-phosphatase (G-6-Pase), protein and DNA content in the liver of the spiny mouse.* On the horizontal axis the age of the animals (in days) is indicated. On the vertical axis enzyme activities are expressed in nmol $\times$ min^{-1} $\times$ mg protein^{-1} at 25°C, except glutamate dehydrogenase that was assayed at 37°C. Protein content is expressed as mg $\times$ g wet weight^{-1}, DNA content as µg $\times$ mg protein^{-1}. No distinction of sex was found. A represents 3–5-month-old adults

phosphatase decrease to 60–70% and 40–50% of the postnatal values respectively. The adult level of tyrosine aminotransferase activity does not differ significantly from the weaning levels, while the adult level of glucose-6-phosphatase activity is approximately 50% of that during the weaning period.

In short, therefore, profiles of urea cycle enzymes can be characterized as two-step profiles with increases in the prenatal and preweaning period, while the profiles of tyrosine aminotransferase and glucose-6-phosphatase are characterized by a monophasic maximum in the first postnatal week. The developmental profile of glutamate dehydrogenase is different from the other developmental profiles. Activity starts to increase 8–10 days before birth and has already reached the developmental maximum 3 days before birth. Activity reaches a plateau during the next 2 weeks and then slowly decreases to 70–80% of the maximum level by 3 weeks after birth.

Protein content of the liver is approximately 40–50% lower before birth than after birth. Protein content may decrease slightly between 12 and 4 days before birth and then increase during the last 3 prenatal and first 3 postnatal days to a plateau of approximately 150 mg $\times$ g wet weight^{-1}.

DNA content of the liver is highest in young fetuses and decreases approximately 80% during the last prenatal week. During the first 2 postnatal weeks DNA content does not change, but from the 3rd postnatal week onwards DNA levels have decreased another 20–30%.

Since hormones have been shown to be important modulating factors of the developmental profiles of liver

Table 1. *Effect of daily hormonal treatments of carbamoyl-phosphate synthetase (CPS), ornithine transcarbamoylase (OTC), arginase (Aase), glucose-6-phosphatase (G-6-Pase) and tyrosine aminotransferase (TAT) activities, 78 h after the start of the treatment on day 1 after birth*
Six animals per treatment were used. At the bottom, the minimum difference between means (%), that is significant at the 5% level [23], is indicated. Values in parentheses give the activity in nmol $\times$ min^{-1} $\times$ mg protein^{-1} (25°C)

Treatment	Activity after 78 h of treatment				
	CPS	OTC	Aase	TAT	G-6-Pase
	% of control				
Control	100 (8.1 ± 0.7)	100 (235 ± 24)	100 (1930 ± 190)	100 (4000 ± 550)	100 (216 ± 40)
Prednisolone	425	145	220	1685	105
Glucagon	155	150	160	1160	115
Triiodothyronine	120	80	105	135	155
Minimum significant difference	20	10	15	140	10

Table 2. *Effect of treatment with dexamethasone acetate and glucagon-Zn-protamine on carbamoyl-phosphate synthetase, ornithine transcarbamoylase and arginase activity in rat and spiny mouse fetuses and neonates, 96 h after the start of the treatment*
Three animals per treatment were used. Results are expressed as -fold stimulation over the control value to indicate effectiveness of the treatment, and as percentages of the adult value to illustrate the difference in inducibility of the enzymes in fetuses and neonates. Control values did not differ significantly from untreated animals, while hormone-treatment values were significantly different from control values ($P < 0.05$) in fetuses and neonates (Student's *t*-test). The coefficient of variation of the different treatment groups was always less than 15%. Adult activities (in mU $\times$ mg protein^{-1} $\times$ min^{-1}) were taken to be 20 for carbamoylphosphate synthetase, 550 for ornithine transcarbamoylase and 4000 for arginase in the rat, and 30, 550 and 3000 for the respective enzymes in the spiny mouse

Enzyme	Activity when sacrificed			
	1 day ante partum		4 days post partum	
	-fold	%	-fold	%
Carbamoyl-phosphate synthetase				
Rat	2.1	8	4.4	106
Spiny mouse	1.4	14	3.7	95
Ornithine transcarbamoylase				
Rat	1.2	40	1.2	91
Spiny mouse	1.2	42	1.5	82
Arginase				
Rat	1.5	28	2.6	85
Spiny mouse	1.3	55	1.8	108

enzymes in the rat [2, 6, 8], we have investigated the effect of three of them in neonatal spiny mouse. Table 1 shows that in these animals prednisolone treatment significantly enhances the level of carbamoyl-phosphate synthetase, ornithine transcarbamoylase, arginase and tyrosine aminotransferase. Glucagon treatment significantly increases the level of carbamoyl-phosphate synthetase, ornithine transcarbamoylase, arginase, tyrosine aminotransferase and glucose-6-phosphatase, while triiodothyronine has a significant stimulatory effect on glucose-6-phosphatase only and an inhibitory effect of ornithine transcarbamoylase.

Table 2 shows the effects of dexamethasone and glucagon treatment on carbamoyl-phosphate synthetase, ornithine transcarbamoylase and arginase activity levels in term rat and spiny mouse fetuses. As spiny mouse dams showed hepatic failure upon repeated administration of ether anesthesia, the dams were treated for 4 days with 30 µg dexamethasone acetate $\times$ g body weight^{-1}. At least in rats such a treatment should result in fetal dexamethasone levels equivalent to a direct injection of 10 µg dexamethasone $\times$ g body weight^{-1} [24]. Glucagon-Zn-protamine complex was used as this complex was shown previously to be more effective in rat fetuses than glucagon alone when administered once daily [8]. For comparison neonatal spiny mice and neonatal rats were injected once daily for 4 days with 10 µg dexamethasone acetate $\times$ g body weight^{-1}, on the 3rd day in combination with 50 µg glucagon-Zn-protamine g body weight^{-1}. The results show that both in rat and in spiny mouse fetuses, carbamoyl-phosphate synthetase, ornithine transcarbamoylase and arginase are clearly inducible by the dexamethasone and glucagon treatment regimen but that the enzyme levels attained are well below the maximally attainable levels directly after birth. The magnitude of this effect is comparable with that of repeated direct administration, at least in rats [8]. If the direct fetal injection with dexamethasone and glucagon complex was repeated once, the stimulation of enzyme activity was not further enhanced in either spiny mice or rats (results not shown).

In order to be able to assess the relation between the parameters of functional development of the liver, i.e. the enzymic profiles, and the morphological development of the liver, Fig. 2 shows the histological architecture of the liver of term fetal rat and spiny mouse. The livers of both species are characterized by vacuolar hepatocytes, containing nuclei with distinct nucleoli (1), haemopoeietic cells that contain either dense nuclei (2a) or polyploid nuclei (2b) and portal (3) and central veins (4). However, term spiny mouse liver contains biliary ducts (5) that accompany the portal veins well into the periphery of its branching pattern. In the rat this condition is only seen several days after birth (not shown). The hepatocytes of term spiny mouse liver are smaller than the hepatocytes of term rat liver possibly indicating a lower glycogen content. It appears unlikely that this observation is

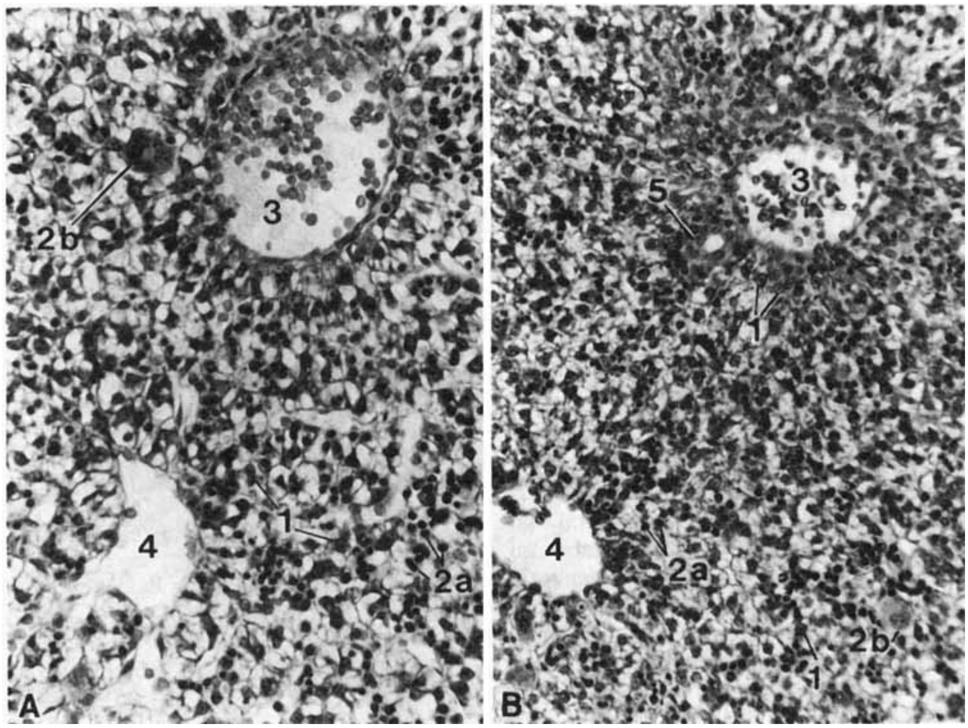

Fig. 2. *Light-microscope photographs of fetal liver tissues on the day of expected delivery (7 μm, haematoxylin and azophloxin stained, × 400).* (A) The rat; (B) the spiny mouse. The hepatocytes (1) appear vacuolar, probably because of the loss of glycogen. Furthermore, many haematopoietic cells are still visible, most of which are characterized by dense nuclei (erythroblasts, 2a), while only a few contain polyploid nuclei (megakaryoblasts, 2b). Note the portal vein (3), the central vein (4) and a biliary duct (5) in the spiny mouse liver only in this peripherial area of the branching pattern of the portal vein

due to shrinkage of the spiny mouse specimen as such differences were not observed in contiguous organs.

DISCUSSION

We have shown previously [5, 16, 17] that in the perinatal period the closely related murinoid species *Rattus norvegicus* and *Acomys cahirinus* have the same developmental time-scale, except for the different developmental timing of birth, resulting in typical altricial (rat) and precocial (spiny mouse) modes of development. A schematic comparison between the enzymic profiles of tyrosine aminotransferase and carbamoyl-phosphate synthetase of rat (derived from [2, 3]) and spiny mouse is shown in Fig. 3. The profiles of tyrosine aminotransferase and carbamoyl-phosphate synthetase were chosen because they are typical for enzymes with perinatal and weaning maxima, respectively. The ages for both species have been aligned in such a way that the developmental stages match [16]. With the presently used assay conditions, enzyme activities are first detectable at approximately 5 (rat) and 13 (spiny mouse) days before birth. At this time both species have reached the same stage of morphological development [5, 25]. This stage of development appears to be important throughout the gastrointestinal tract for the acquisition of the capacity to synthesize enzymes at detectable rates [17]. For the pancreas this developmental stage has been called the secondary transition, i.e. from the protodifferentiated to the differentiated, mature state in which enzyme synthesis can be modulated by hormones [26]. These common developmental transitions in rat and spiny mouse therefore underscore the common developmental time-scale in both species.

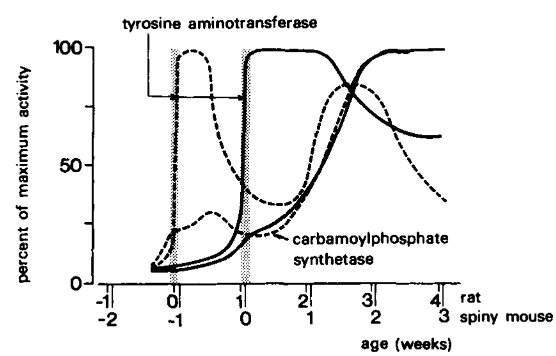

Fig. 3. *Schematic developmental profiles of rat (– – – –) and spiny mouse (———) carbamoyl-phosphate synthetase and tyrosine aminotransferase.* On the horizontal axis the ages for both species have been aligned in such a way that developmental stages match [16]. Birth is indicated by 0. Note that the birth-associated phenomena, which are highlighted by tyrosine aminotransferase, occur approximately 8 developmental days apart but that the weaning-associated phenomena, which are highlighted by carbamoyl-phosphate synthetase, occur at the same developmental stage. Note furthermore that the developmental changes of the rat are typically biphasic, while those of the spiny mouse are typically monophasic. The rat profiles for carbamoyl-phosphate synthetase and tyrosine aminotransferase were obtained from [2, 3]

Tyrosine aminotransferase (Fig. 3) shows the typical developmental profile of an enzyme belonging to the perinatal cluster [1] in both species, with a developmental maximum shortly after birth, i.e. 7–8 developmental days later in the spiny mouse than in the rat. However, a major difference between both species is the presence of only a single peak in enzymic activity in the spiny mouse and two peaks in activity (perinatal and weaning) in the rat. It would appear that in the spiny mouse the two peaks of enzyme activity, that are observed in the rat, have merged into a single peak of activity. The developmental profile of carbamoyl-phosphate synthetase shows, in the rat, rises in enzyme activity levels in the perinatal and weaning period and, similar to tyrosine aminotransferase, relatively low activity in between, i.e. in the 2nd postnatal week. In the spiny mouse a moderate perinatal increase is also observed, and again, similar to tyrosine aminotransferase, this perinatal phase merges with the subsequent, larger increase of the 2nd postnatal week.

The picture therefore emerges that irrespective of the mode of development, many organotypic enzymes first become enzymatically detectable upon the transition from the protodifferentiated to the differentiated state [26], i.e. at the transition from the embryonic to the fetal period. Furthermore, both altricial and precocial species have perinatal and weaning enzyme clusters. While the maximum of the perinatal phase of enzymic profiles is dictated by the developmental timing of birth (and associated hormonal adaptation) and therefore 7–8 developmental days apart in rat and spiny mouse, the surge of the weaning phase of enzymic profiles may again follow an inherent biological program (with additional, superimposed hormonal modulation) because the developmental profiles of organotypic enzymes coincide during this period in both species, also in other parts of the gastrointestinal tract [5, 17, 18]. Furthermore, it has only been possible to modify, not abolish these increases in enzyme activity, at least in the rat [3, 27]. The main difference between rat and spiny mouse, therefore, resides in the different developmental timing of birth, resulting in well-separated perinatal and weaning phases of enzymic profiles in the rat and merged perinatal and weaning phases in the spiny mouse owing to 'delayed' occurrence of birth. In this respect the observation, that the morphological development of the liver (Fig. 2) proceeds independent of the timing of birth and reflects the altricial or precocial condition at birth, is interesting.

The previous discussion raises the question of the regulatory factors involved in the establishment of the respective enzymic profiles. The identity of these factors in the perinatal period is especially relevant to establish the 'spiny mouse model' as complementary to the 'rat model' (cf. [14]). From Table 1 it is clear that glucocorticosteroids, cyclic AMP and thyroid hormone can modulate the level of all enzymes tested. Furthermore, comparison with previous data on the rat [2, 6, 8] shows that the effects of the modulating factors are the same in rat and spiny mouse.

Since the late intra-uterine developmental stages of the spiny mouse [5, 16, 17], including its liver (Fig. 2), seem comparable to the early extra-uterine developmental stages of the rat, the change of limited to complete inducibility of organotypic gene expression at birth in both species (Tables 1, 2 [2, 6–8]) suggests that the conditions of the intra-uterine milieu rather than the more or less advanced state of development of the respective parenchymal cells is responsible. It has been shown in rat [7] that the simultaneous presence of estrogen and progesterone represents such an inhibiting condition of the late fetal intra-uterine milieu.

REFERENCES

1. Greengard, O. (1970) in *Biochemical Actions of Hormones*, vol. 1 (Litwack, G., ed.) pp. 53–87, Academic Press, New York.
2. Lamers, W. H. & Mooren, P. G. (1980) *Biol. Neonate 37*, 113–137.
3. Lamers, W. H. & Mooren, P. G. (1981) *Mech. Ageing Dev. 15*, 77–92.
4. Henning, S. J. (1981) *Am. J. Physiol. 241*, G199–G214.
5. Lamers, W. H., Mooren, P. G., Oosterhuis, W. P., Lunstroo, H., de Graaf, A. & Charles, R. (1982) *Adv. Exp. Med. Biol. 153*, 229–240.
6. Lamers, W. H. & Mooren, P. G. (1980) *Biol. Neonate 37*, 264–284.
7. Lamers, W. H. & Mooren, P. G. (1981) *Biol. Neonate 40*, 78–90.
8. Lamers, W. H. & Mooren, P. G. (1981) *Mech. Ageing Dev. 15*, 93–118.
9. Lamers, W. H. & Mooren, P. G. (1981) *Mech. Ageing Dev. 15*, 119–128.
10. Greengard, O., Sahib, M. K. & Knox, W. E. (1970) *Arch. Biochem. Biophys. 137*, 477–482.
11. Greengard, O. & Dewey, H. K. (1971) *Proc. Natl Acad. Sci. USA 68*, 1698–1701.
12. Gautier, C., Husson, A. & Vaillant, R. (1977) *Biochimie (Paris) 59*, 91–95.
13. Husson, A. & Vaillant, R. (1982) *Endocrinology 110*, 227–232.
14. Greengard, O. (1977) *Pediatr. Res. 11*, 669–676.
15. Jones, C. T. (1982) *The Biochemical Development of the Fetus and Neonate*, Elsevier Biomedical Press, Amsterdam.
16. Oosterhuis, W. P., Mooren, P. G., Charles, R. & Lamers, W. H. (1984) *Biol. Neonate 45*, 236–243.
17. Lamers, W. H., Mooren, P. G. & Charles, R. (1984) *Biol. Neonate*, in the press.
18. Diamonstone, T. I. (1966) *Anal. Biochem. 16*, 395–401.
19. Schmidt, E. (1974) in *Methods of Enzymic Analysis* (Bergmeyer, H. U., ed.) pp. 650–656, Verlag Chemie, Weinheim.
20. Baginski, E. S., Foà, P. P. & Zak, B. (1974) in *Methods of Enzymic Analysis* (Bergmeyer, H. U., ed.) pp. 876–880, Verlag Chemie Weinheim.
21. Cleland, K. W. & Slater, E. C. (1953) *Biochem. J. 53*, 547–556.
22. Burton, K. (1956) *Biochem. J. 62*, 315–323.
23. Dunnett, C. W. (1955) *J. Am. Stat. Ass. 50*, 1096–1121.
24. Funkhouser, J. D., Peevey, K. J., Mockridge, P. B. & Hughes, E. R. (1978) *Pediatr. Res. 12*, 1053–1056.
25. Dieterlen, F. (1962) *Z. Säugetierkd. 28*, 193–227.
26. Rutter, W. J., Kemp, J. D., Bradshaw, W. S., Clark, W. R., Ronzio, R. A. & Sanders, T. G. (1968) *J. Cell. Physiol. 72*, suppl. 1, 1–18.
27. Blake, H. H. & Henning, S. J. (1983) *Am. J. Physiol. 244*, R537–R543.

W. H. Lamers, P. G. Mooren, A. de Graaf, and R. Charles, Anatomisch Embryologisch Laboratorium, Academisch Medisch Centrum, Meibergdreef 15, NL-1105-AZ Amsterdam, The Netherlands